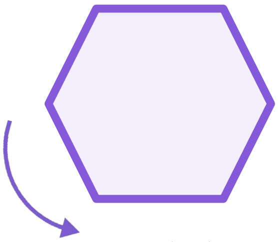
Describing a rotation's direction and size

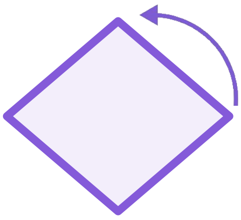
Task A: Describing the direction of a rotation

1) In which direction does this arrow suggest the rotation is happening?



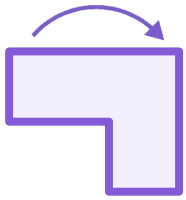
anti-clockwise

2) In which direction does this arrow suggest the rotation is happening?



anti-clockwise

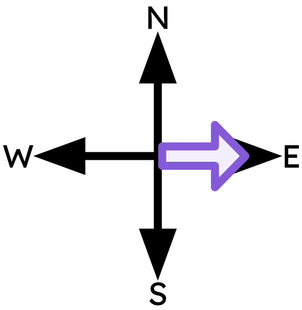
3) In which direction does this arrow suggest the rotation is happening?



clockwise

4) An arrow is pointing E (east).

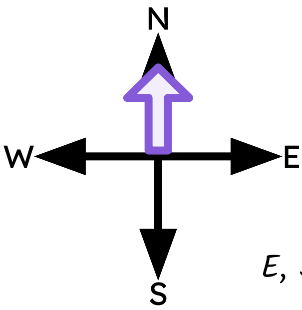
If the arrow rotates anti-clockwise, which compass direction will it pass first?



N (north)

5) An arrow is pointing N (north).

The arrow will rotate clockwise until it points N (north) again. In what order will it pass through the other compass directions?

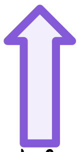


E, S, W

Task B: Describing the size of a rotation

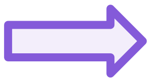
1) Write down the direction (up, down, left, right) each arrow will be pointing in after the rotation has been applied.

a) 90° rotation anti-clockwise



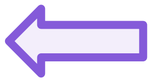
left

b) 180° rotation clockwise



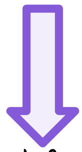
left

c) 270° rotation anti-clockwise



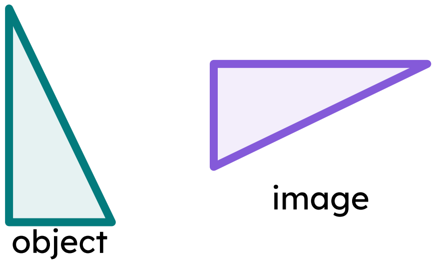
up

d) 90° rotation clockwise



left

2) The object has been rotated by less than 360° to create its image. Complete the two sentences that could describe this transformation.



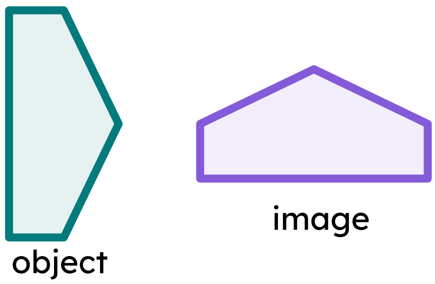
Possible Statement 1:

The object has been rotated *90°* clockwise.

Possible Statement 2:

The object has been rotated 270° *anti-clockwise*.

3) The object has been rotated by less than 360° to create its image. Complete the two sentences that could describe this transformation.



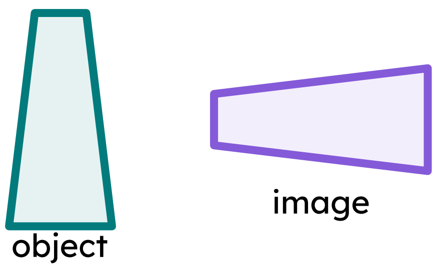
Possible Statement 1:

The object has been rotated 90° *anti-clockwise*.

Possible Statement 2:

The object has been rotated *270°* *clockwise*.

4) The object has been rotated by less than 360° to create its image. Complete the two sentences that could describe this transformation.



Possible Statement 1:

The object has been rotated *270°* clockwise.

Possible Statement 2:

The object has been *rotated* *90°* anti-clockwise.

Task C: Describing more complex sizes of rotation

1) Write down the calculation and single **rotation** angle in degrees to each of the following statements.

a) between a quarter turn and a half turn

between 90° and 180° = 135°, or any number between 90° and 180°.

b) two full turns

360° + 360° = 720°

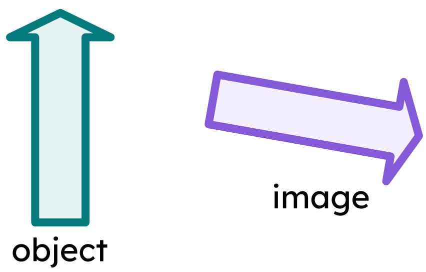
c) between no **rotation** and a quarter turn

between 0° and 90° = 45°, or any number between 0° and 90°.

d) one and three-quarter turns

360° + 270° = 630°

2) The object has been rotated by less than 360° to create its image.
Complete the two sentences that could describe this transformation.



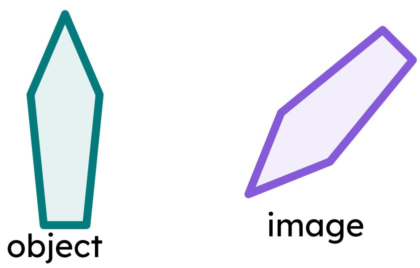
Possible Statement 1:

The object has been rotated 100° clockwise.

Possible Statement 2:

The object has been rotated *260°* anti-clockwise.

3) The object has been rotated by less than 360° to create its image.
Complete the two sentences that could describe this transformation.



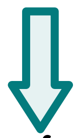
Possible Statement 1:

The object has been rotated 140° *anti-clockwise*.

Possible Statement 2:

The object has been rotated *220°* *clockwise*.

4) This arrow will be rotated anti-clockwise through two full turns and three-quarters of another.



a) Which direction will the image of the arrow be pointing?

It will be facing left.

b) In total, how many degrees will the arrow have rotated?

360° + 360° + 270° = 990°

c) Give a description of a different anti-clockwise rotation that would result in the arrow pointing in the same direction.

Any number of full turns (including zero) and three-quarters of another.

d) Give a description of a clockwise rotation that would result in the arrow pointing in the same direction.

Any number of full turns (including zero) and one-quarter of another.